1.4.2 Express each of the signals in Fig. 1 by a single expression valid for all t.

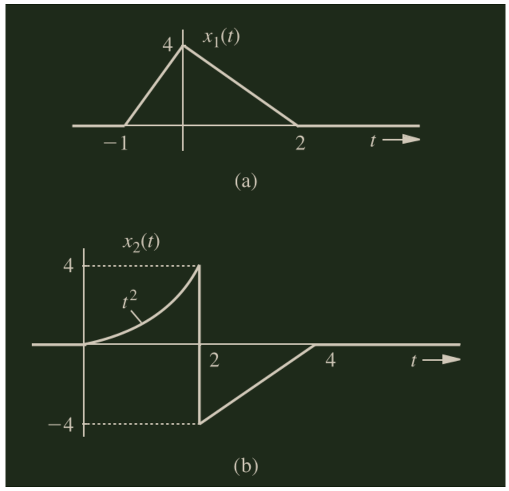

Figure 1: CT signals

a) Using point-slope form, $(y - y_1) = m(x - x_1)$, we can express (a) as a piecewise function, where the first section can be expressed as $(y-4) = 4(x-0) \rightarrow x_{1a}(t) = 4t+4$, and the second section can be expressed as $(y-4) = -2(x-0) \rightarrow x_{1b}(t) = -2x+4$. More succinctly,

$$x_1(t) = \begin{cases} 4t+4 & -1 < t < 0 \\ -2t+4 & 0 < t < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Using the unit step function,

$$(4t+4)[u(t+1) - u(t)] + (-2t+4)[u(t) - u(t-2)].$$

b) The first section of (b) is defined already as t^2, and the second section may be found in point-slope form as $(y+4) = 2(x-2) \rightarrow x_{2b}(t) = 2t-8$. More succinctly,

$$x_2(t) = \begin{cases} t^2 & 0 < t < 2 \\ 2t-8 & 2 < t < 4 \\ 0 & \text{elsewhere.} \end{cases}$$

Using the unit step function,

$$t^2[u(t) - u(t-2)] + (2t-8)[u(t-2) - u(t-4)].$$

1.4.4 Define signal $x(t) = u(t-1) - u(t-2.5) - 2\delta(t-4) + \delta(t-6)$.

a) Sketch $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$.

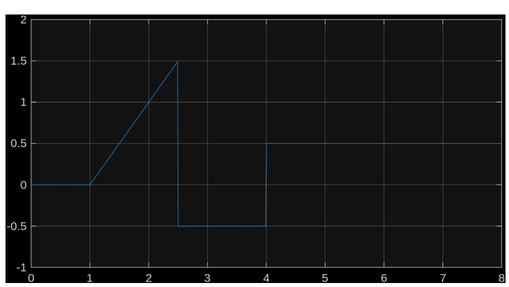

Figure 2: Plot of $y(t)$

b) Describe a simple change that can be made to the right-most delta function in $x(t)$ so that $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$ has finite energy. Setting the coefficient of $\delta(t)$ to 0.5 will "zero out" the function after $t = 4$.

c) Sketch $z(t) = \int_{t}^{\infty} x(\tau)\, d\tau$.

d) Determine real constants A and B so that $w(t) = x(\frac{t-A}{B})$ has a region of support $[-2, 2]$.

1.7.1 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, determine which of the systems are linear and which are nonlinear.

a)

$$\frac{dy(t)}{dt} + 2y(t) = x^2(t).$$

Testing the additive property first,

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1(t)^2$$
$$\frac{dy_2(t)}{dt} + 2y_2(t) = x_2(t)^2.$$

Then,

$$\frac{d}{dt}[k_1y_1(t) = k_2y_2(t)] + 2[k_1y_1(t) + k_2y_2(t)] = k_1x_1(t)^2 + k_2x_2(t)^2.$$

2.2.2 Consider a linear time-invariant system with input $x(t)$ and output $y(t)$ that is described by the differential equation

$$(D + 1)(D^2 - 1)y(t) = (D^5 - 1)x(t)$$

Furthermore, assume $y(0) = \dot{y}(0) = \ddot{y}(0) = 1$.

(a) What is the order of this system?

(b) What are the characteristic roots of this system?

(c) Determine the zero-input response $y_{\text{zir}}(t)$. Simplify your answer.

2.2.3 A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D)y(t) = (2D^3 + 1)x(t).$$

(a) What is the characteristic equation of this system?

(b) What are the characteristic modes of this system?

(c) Assuming $y_{\text{zir}}(0) = 4$, $\dot{y}_{\text{zir}}(0) = -18$, and $\ddot{y}_{\text{zir}}(0) = 0$, determine this system's zero-input response $y_{\text{zir}}(t)$. Simplify $y_{\text{zir}}(t)$ to include only real terms (i.e., no j's should appear in your answer).

2.2.11 A system is described by a constant-coefficient linear differential equation and has zero-input response given by $y_0(t) = 2e^{-t} + 3$.

(a) Is it possible for the system's characteristic equation to be $\lambda + 1 = 0$? Justify your answer.

(b) Is it possible for the system's characteristic equation to be $\sqrt{3}(\lambda^2 + \lambda) = 0$? Justify your answer.

(c) Is it possible for the system's characteristic equation to be $\lambda(\lambda + 1)^2 = 0$? Justify your answer.

2.3.5 Find the unit impulse response of an LTIC system specified by the equation specified by the equation

$$(D^2 + 6D + 9)y(t) = (2D + 9)x(t).$$

2.3.7 A causal LTIC system with input $x(t)$ and output $y(t)$ is described by the constant coefficient integral equation

$$y(t) + \int 3y(t)dt + \int\int 2y(t)dt = \int\int x(t)dt - \int\int\int x(t)dt.$$

(a) Express this system as a constant coefficient linear differential equation in standard operator form.

(b) Determine the characteristic modes of this system.

(c) Determine the impulse response $h(t)$ of this system.

2.3.4 Find the unit impulse response for the first-order allpass filter specified by the equation

$$(D+1)y(t) = -(D-1)x(t)$$

2.4.13 Using direct integration, find $e^{-at}u(t) * e^{-bt}u(t)$.

2.4.16 The unit impulse response of an LTCI system is

$$h(t) = e^{-t}u(t)$$

Find this system's (zero-state) response $y(t)$ if the input $x(t)$ is

a) $u(t)$

b) $e^{-t}u(t)$

c) $e^{-2t}u(t)$

d) $\sin(3t)u(t)$

Use the convolution table (Table 2.1) to find your answers.

2.4.21 A first order allpass filter impulse response is given by

$$h(t) = -\delta(t) + 2e^{-t}u(t)$$

a) Find the zero-state response of this filter for the input $e^t u(-t)$.

b) Sketch the input and the corresponding zero-state response.

2.4.23 The zero-state response of an LTIC system to an input $x(t) = 2e^{-2t}u(t)$ is $y(t) = [4e^{-2t} + 6e^{-3t}]u(t)$. Find the impulse response of the system. [*Hint:* We have not yet developed a method of finding $h(t)$ from the knowledge of the input and the corresponding output. Knowing the form of $x(t)$ and $y(t)$, you will have to make the best guess of the general form of $h(t)$.]